Requirements Specification & Use Cases

Product Requirements Document (PRD), Functional & Non-Functional Specifications, and Formal Use Cases • IEEE Std 830-1998 Aligned

1. System Scope & Objectives

Umkhuseli is a dual-persona Android and Web application designed to mitigate the acute physical risks faced by South African survivors of Gender-Based Violence (GBV). The software provides three core services: **1) Covert Disguise & Panic Camouflage** (PowerCalc SA prepaid electricity utility and store-themed launcher), **2) Hardware-Encrypted Legal Evidence Vault**, and **3) Zero-Latency / Zero-Airtime Emergency SOS Beaconing**. All requirements conform to the statutory standards of the **South African Domestic Violence Act (Act 116 of 1998, amended by Act 14 of 2021)**, the **Protection of Personal Information Act (POPIA Act 4 of 2013)**, and the **Electronic Communications and Transactions Act (ECTA Act 25 of 2002)**.

2. Functional Requirements Matrix (FR)

Req ID	Feature Module	Functional Specification & Acceptance Criteria	Priority
FR-01	PowerCalc SA Camouflage Facade	Default UI acts as a functional municipal prepaid electricity and STS token calculator (Ekurhuleni/Boksburg, City Power, Tshwane, Cape Town, Eskom) or store-themed launcher.	**MANDATORY**
FR-02	Covert Safe Haven Triggers	Unlocks Safe Haven via mathematical PIN sequence (1234=, 8426=, 0000=) or STS Token button long-press on calculator keypad.	**MANDATORY**
FR-03	Biometric Hardware Gate	Integrates with Android BiometricPrompt (BIOMETRIC_STRONG) verifying Face, Fingerprint, or secure device PIN before vault entry.	**MANDATORY**
FR-04	AES-256-GCM Evidence Diary	Encrypts descriptions, injuries, and witness statements at-rest in Room SQLite with AndroidKeyStore keys and unique 12-byte IVs.	**MANDATORY**
FR-05	0ms Camouflage Panic Wipe	Pressing hardware Back, Escape, or Quick-Exit immediately scrubs decrypted in-memory variables and restores PowerCalc SA in 0 ms.	**CRITICAL**
FR-06	Tactile SOS Grace Dispatcher	State machine executes 3-second tactile vibration countdown; user can cancel before dispatch to prevent false alarms.	**MANDATORY**
FR-07	72-Hour GPS Beacon	Broadcasts live coordinates, battery percentage, charging state, handset model, and Find My Device tracker URL via SMS to 5 ICE contacts.	**MANDATORY**
FR-08	Zero-Airtime Lifelines	One-touch links to SAPS 10111, toll-free GBVCC 0800 428 428, and zero-rated USSD callback pings (*120*7867# / *134*7355#).	**MANDATORY**
FR-09	Thuthuzela Care Directory	Offline-cached directory of accredited TCC forensic centers covering all 9 SA provinces with calling and navigation coordinates.	**MANDATORY**
FR-10	Form 2 Protection Dossier	Compiles timestamped incidents into South African Form 2 affidavits with Section 4(2) safe address withholding right.	**MANDATORY**
FR-11	Duress PIN Deception (9999=)	When coerced, entering PIN 9999= displays an authentic municipal electricity recharge voucher while silently dispatching SOS telemetry.	**CRITICAL**
FR-12	UAT Field Testing & Intercept	Dedicated UAT toggle intercepting SOS dispatch to simulated test guardians (+27 82 000 0001/2) with in-app POPIA feedback form and offline outbox.	**OPERATIONAL**
FR-13	Cross-Platform Vercel PWA	Vercel-hosted SPA for desktop PC/Mac and iOS Safari zero-trace access, plus direct APK downloads from hosted domain.	**MANDATORY**

3. Non-Functional Requirements Matrix (NFR)

NFR ID	Dimension	Standard / Metric	Implementation Architecture
NFR-01	Camouflage Latency	≤ 0 ms perceptual delay on panic exit; 0 traces in task manager.	Synchronous ViewModel state wipe; immediate recomposition to PowerCalc SA.
NFR-02	Cryptographic Strength	256-bit AES-GCM; authenticated encryption with 128-bit MAC tag.	Hardware StrongBox / TEE via AndroidKeyStore; Web SubtleCrypto AES-GCM.
NFR-03	Data Privacy (POPIA)	Zero third-party trackers; local-first storage; zero telemetry leakage.	Pure local Room SQLite / Dexie IndexedDB with zero-knowledge encryption.
NFR-04	Zero-Data Availability	100% core life-safety features operational with R0.00 data / airtime.	SAPS 10111, GBVCC toll-free line, and zero-rated USSD *120*7867#.
NFR-05	Cross-Platform Deploy	Android 7.0-16 APK; Desktop PC/Mac & iOS Safari responsive PWA.	Jetpack Compose BOM + React 19 / TypeScript hosted on Vercel edge.

4. Core Formal Use Cases

UC-01: Covert Safe Haven Unlock & Biometric Challenge

Actors: Survivor (Primary)
Preconditions: App running in PowerCalc SA camouflage facade mode.
Trigger: User enters secret calculation (1234=, 8426=, or 0000=) or long-presses STS Token button.
Main Flow: 1. PowerCalc captures trigger sequence without audio/visual feedback. 2. Prompts BiometricPrompt challenge. 3. User authenticates via Fingerprint or Face. 4. Master key retrieved from AndroidKeyStore. 5. Safe Haven screen rendered with encrypted vault diary.
Alternative Flow: If duress PIN 9999= entered, execute UC-07 (Duress Deception & Silent Dispatch).
Postconditions: Vault unlocked; memory session active.

UC-02: Armed SOS Emergency Distress Broadcast

Actors: Survivor, ICE Guardians (5 contacts), SAPS Emergency Flying Squad
Preconditions: At least one active ICE guardian configured with mobile number.
Trigger: User presses Emergency SOS button.
Main Flow: 1. SafetyTriggerViewModel transitions to Armed state. 2. 3-second tactile countdown begins with distinct haptic pulses. 3. Grace timer expires. 4. Device queries last known GPS coordinates and battery telemetry. 5. Persists distress incident to Room DB. 6. Transmits 72-hour beacon SMS to all active guardians.
Alternative Flow: User taps Cancel during 3-second countdown; state resets to Cancelled; no SMS sent.
Postconditions: SOS dispatched; incident logged to legal vault.

UC-03: 0-Millisecond Panic Camouflage Scrub

Actors: Survivor, Threat Actor (Physical Inspector)
Preconditions: Safe Haven vault is actively open.
Trigger: User presses hardware Back button, Escape key, or top panic bar.
Main Flow: 1. Event handler intercepts event synchronously. 2. In-memory decrypted strings and cryptographic pointers zeroed immediately. 3. PowerCalc SA utility instantly displayed in 0 ms. 4. Recents task thumbnail blanked.
Postconditions: No safety traces present; device displays innocent municipal electricity calculator.

UC-04: Live Device UAT Simulated Testing & In-App Feedback

Actors: UAT Field Tester, Quality Assurance Lead
Preconditions: UAT Testing Mode toggled ON in Safe Haven dashboard.
Trigger: Tester triggers Quick SOS or completes test scenario checklist.
Main Flow: 1. System intercepts SOS dispatch to simulated test guardians (+27 82 000 0001/2). 2. Renders telemetry card showing mock SAPS radio dispatch. 3. Tester navigates to Feedback tab and submits usability rating. 4. Record saved to local outbox and synced to Supabase `uat_feedback_submissions`.

Postconditions: Live testing validated without real emergency escalation; POPIA-compliant telemetry logged.